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Linear Algebra

Linear structure ⟨ $f(a+b) = f(a) + f(b)$; $f(sx) = sf(x)$

LI/LD → $s_1v_1 + s_2v_2 + \cdots + s_nv_n = 0$ } If ∃ some non-zero s_i → LD; only soln is all $s_i = 0$ → LI

Row operations ⟨
- $R_i \leftrightarrow R_j$ swap
- $R_i \leftarrow kR_i$ scalar multiplication
- $R_i \leftarrow R_i \pm kR_j$ linear comb^n (makes matrix non-invertible (rows LD))

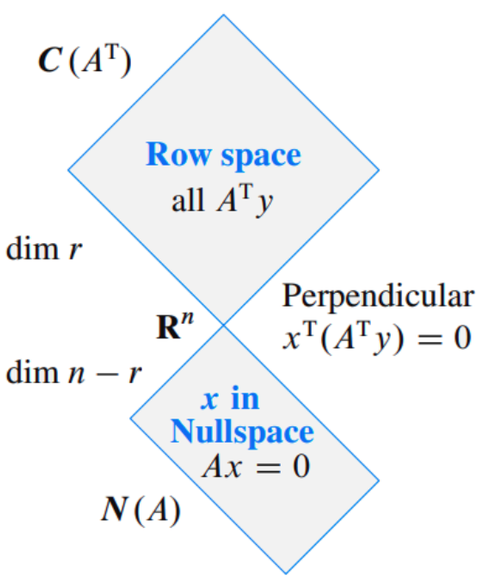
↳ Do not affect the row space or null space

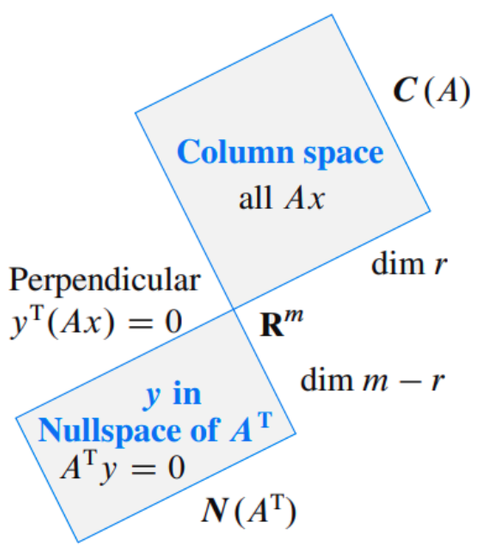
Vector space : lin. comb^{n}s of vectors are still in space

To check subspace → zero vector ($\vec{0}$)
→ additive inverse.
→ closure over (+), (·).

NOTE:
$\dim(V_1 \cup V_2) = \dim(V_1) + \dim(V_2) - \dim(V_1 \cap V_2)$

Basis : minimal set (LI) that spans a vector space. ϕ (Basis) = dimension.





Full Column Rank	Full Row Rank	Full Rank
$r = n \leq m$ (Tall/Sq)	$r = m \leq n$ (Wide/Sq)	$r = m = n$ (Sq)
Null Space: $\{\vec{0}\}$	**Left Null Space**: $\{\vec{0}\}$	Both have only the zero vector.
All rows are LI	All columns are LI	Both are LI
Solutions: 0/1 solution	**Solutions**: ∞ solutions	**Solutions**: 1 solution (Unique solution)

$A_{m\times n}\, x_{n\times 1} = b_{m\times n}$
(x: null space; b: col^m space)

Echelon form → $\left(\frac{n^2-n}{2}\right)$ operations
RREF → n^2 operations
} $O(n^2)$

$Ax = 0$
- unique soln → trivial soln → A has LI col^{m}s. → $r = n$
- ∞ soln → A has LD col^{m}s. → $r < n$

$Ax = b$ — $b \notin col(A)$ → No soln → $rank(A) < Rank(A|b)$ → $b \notin col(A)$

$b \in col(A)$; $rank(A) = rank(A|b) = r$
- unique soln → $r = n$ → no free vars → $N(A) = \{\vec{0}\}$ → LI col^{m}s.
- ∞ soln → $r < n$ → LD col^{m}s. (causes ∞ soln^s.)

$x = x_p + x_n$

After converting to RREF:

$$\begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -F \\ I \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

(soln)

* First check soln exists/not → row rank
Then check if col^{m}s are LD/LI
- LD → ∞ soln^s
- LI → 1 soln(s)

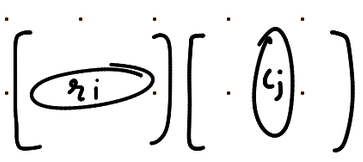
$A \xrightarrow{E} R$

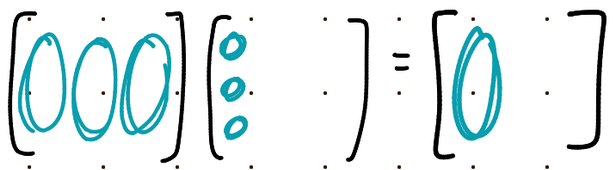
$\dim(\text{null space}) = \#$ of dep col^{m}s

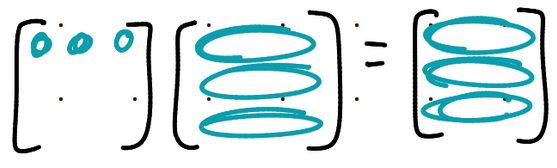
$\dim(\text{left null space}) = \#$ of dep rows

$\dim(\text{row space}) = \#$ of non-zero rows in R $= \dim(\text{col}^m \text{ space})$

Matrix multiplication

a_{ij} = 





* When a matrix is multiplied by an invertible matrix, its rank remains.

rank(A * some invertible matrix) = rank(A)

To find expr like $B^{-1}b$, solve for $Bx=b$. (Esp. when B is upper/lower Δar)

$AB=C$

If cols of B are LD ⇒ cols of C will be LD.

$Bx=0 \ \exists x \neq 0$

$ABx=0$

$Cx=0$

For col. of C = AB to have LI cols, A and B should have LI cols.

* If A is some scalar multiple of I ($A=\lambda I$) then $AB=BA \ \forall B$.

$AB=0$

↳ cols of B are in N(A)

↳ rows of A are in $N(B^T)$

LU Decomposition

$EA=U$

$A=LU$

For permutation matrix, $P^T=P^{-1}$. (orthogonal)

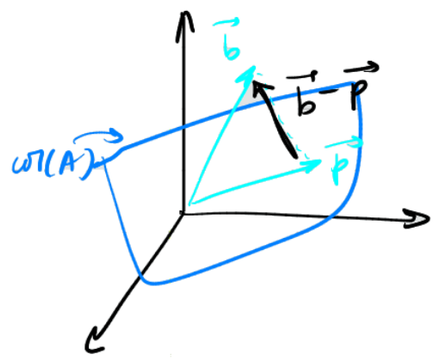
Projections

$Ax = b$ Instead solve $A\hat{x} = p$

$A^TA\hat{x} = A^Tb$

$\vec{p} = A\hat{x} = \underbrace{A(A^TA)^{-1}A^T}_{\text{P (projection matrix)}}b$

- $P = A(A^TA)^{-1}A^T$
- $col(P) = col(A)$
- $rank = 1$
- $P^2 = P$
- $P^T = P$



For projecting onto a vector $v \longrightarrow P = \frac{vv^T}{\|v\|_2^2}$

* If P projects onto a subspace, $(I-P)$ projects onto its orthogonal complement.

Orthogonal Matrix

$\begin{bmatrix} | & & | \\ q_1 & \cdots & q_n \\ | & & | \end{bmatrix}$ $q_i^Tq_i = 1$ } unit length $q_i^Tq_j = 0$ } orthogonal } orthonormal colm vectors

$Q^TQ = QQ^T = I$

↳ only for sq.

If square, orthogonal colms $\Rightarrow$ invertible.

Projection on orthogonal matrix: $\vec{p} = P\vec{b} = QQ^T\vec{b}$

Transformations

linear $\equiv$ matrix exists. $T(v) = Av$

$T(\alpha v_1 + \beta v_2) = \alpha T(v_1) + \beta T(v_2)$

change of basis (B_1 to B_2):

convert B_1 to std basis $(B_1 v) \longrightarrow$ convert to B_2 $(B_2^{-1}B_1 v)$

Eigen Stuff

$Ax = \lambda x$ solve for λ: $|A - \lambda I| = 0$

- sum of eigenvalues = Trace(A)
- Product of eigenvalues = Det(A)

- Elimination does NOT preserve eigenvalues.
- If sum of every col^m = 1 $\Rightarrow$ one of the eigenvalues is 1.

Algebraic multiplicity: # of times λ repeats.
Geometric multiplicity: # of LI eig. vecs. corresponding to a λ value.

GM $\leq$ AM

	Eigenvalues	Eigenvectors
Symmetric/ Hermitian	real	orthogonal
Skew-Symm.	zero/imaginary	orthogonal
orthogonal	real/img $\lvert\lambda\rvert = 1$.	orthogonal

If eig. val for A is λ, then eig. val. for $f(A) = f(\lambda)$.

e.g. $A^2 \rightarrow \lambda^2$
$A + 9I \rightarrow \lambda + 9$
$(A + A^{-1}) \rightarrow \lambda + \frac{1}{\lambda}$

- For Dau matrices, λs are the diagonal entries.
- If $\lambda = 0$ is repeated K times $\longrightarrow$ Nullity = K.

$$n - K \leq \text{rank} \leq n - 1$$

- Real symmetric matrices $\rightarrow$ exactly K LI eig vecs for λ if λ is repeated k times.
- Every matrix satisfies its characteristic eqn.
- AB and BA have the same non-zero eigen values. (all same for sq)
- If an eig. val is imaginary $\rightarrow$ conjugate is also eigen val. (-i in eig vec as well).

Positive Definite matrices

- Only +ve eigen values and symmetric. $\lambda > 0$
- For all $x \neq \vec{0}$, $x^T A x > 0$
- A^TA and AA^T are PD.
- All leading determinants > 0.
- All pivots > 0

$\geq 0 \rightarrow$ PSD
$< 0 \rightarrow$ ND
$\leq 0 \rightarrow$ NSD

Quadratic form:

$$ax_1^2 + bx_1x_2 + cx_2^2$$
$$x^TAx$$

$$A = \begin{array}{c} \\ x_1 \\ x_2 \end{array} \begin{array}{c} \begin{array}{cc} x_1 & x_2 \end{array} \\ \begin{bmatrix} a & b_1 \\ b_2 & c \end{bmatrix} \end{array}$$

$b_1 + b_2 = c$

For PD:
$b_1 = b_2 = b/2$
$x^TAx > 0$

A is PD $\Rightarrow$ A^{-1} is PD.
A, B are PD $\Rightarrow$ A+B is PD

Similar matrices : $AS = SB \Rightarrow A = SBS^{-1}$

- same rank and same char. polynomial. → ∴ same eigenvalues

Diagonalization

$A = S\Lambda S^{-1}$

↑ eig. vals ↑ eig. vecs

- For symmetric A, $A = Q\Lambda Q^T$
- $A^k = S\Lambda^k S^{-1}$

Singular Value Decomposition :

$$A = U\Sigma V^T$$
$$m\times n \quad m\times m \; m\times n \; n\times n$$

$$A \begin{bmatrix} | & & | & | & & | \\ v_1 & \cdots & v_r & v_{r+1} & \cdots & v_n \\ | & & | & | & & | \end{bmatrix} = \begin{bmatrix} | & & | & | & & | \\ u_1 & \cdots & u_r & u_{r+1} & \cdots & u_m \\ | & & | & | & & | \end{bmatrix} \begin{bmatrix} \sigma_1 & & & & \\ & \ddots & & & \\ & & \sigma_r & & \\ & & & 0 & \\ & & & & \ddots \end{bmatrix}$$

row space | null space | col space | left null space

σ^2 = eig. vals of $AA^T = A^TA$

V = eig. vecs. of A^TA

U = eigenvecs. of AA^T

$u_i = \frac{1}{\sigma_i} Av_i$ → to find extra u_is, find vecs in left null space (A).

- Singular values of $A^{-1} \rightarrow \frac{1}{\sigma}$
- For symmetric matrices → $\sigma = |\lambda|$ λ of A
- Projection matrix → For C(A): UU^T
  C(A^T): VV^T

Rank =

- rank(A) = # of non-zero singular values of A.
- For orthogonal matrices :
  $A^TA = I$
  $V\Sigma^2 V^T = I$
  singular values = 1

$$A \begin{bmatrix} | & & | \\ v_1 & \cdots & v_r \\ | & & | \end{bmatrix}_{n\times r} = \begin{bmatrix} | & & | \\ u_1 & \cdots & u_r \\ | & & | \end{bmatrix}_{m\times r} \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix}_{r\times r}$$

} reduced form

Matrices

$A^2 = A \rightarrow$ Idempotent

$A^K = 0 \rightarrow$ Nilpotent

$A = A^T \rightarrow$ symmetric

$A = -A^T \rightarrow$ skew-symmetric (diagonal zero)

$A = \bar{A}^T \rightarrow$ Hermitian

$A^2 = I \rightarrow$ involutary

$Q^T Q = Q Q^T = I \rightarrow$ Orthogonal

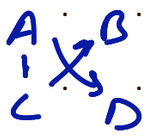
Idempotent matrix:

- $A^2 = A$
- eig. val $\rightarrow$ 0, 1
- eig. vecs are LI
- rank = trace
- Diagonalizable

Block matrices

- For addition: $(A+B)$ shape of matrices is same; Partitioning is same.
- For matrix multiplication: (AB) partition in cols of A = partition in rows of B; check compatibility of A, B first $A_{mxn}\ B_{nxr}$
- Determinants:
  - For diagonal/bar matrices $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$ $\det = \det(A) \cdot \det(D)$
  - commutative $AC = CA$: $\det = \det(AD - CB)$

    $BD = DB$: $\det = \det(DA - BC)$

  - $\begin{pmatrix} A & B \\ B & A \end{pmatrix}$ $\det = \det(A-B)(A+B)$

—

  - transpose

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix}^T = \begin{pmatrix} A^T & C^T \\ B^T & D^T \end{pmatrix}$$

  - Eigen stuff:

$$M = \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} \qquad \text{eigenvalues of } m = \{\text{eigvals of } A, \text{eigvals of } D\}$$

Determinants : area/volm bounded by colm vectors (2D/3D)

(amount of change a matrix can bring to a sq of unit area)

- Non-square transformation → det = 0.
- row swap or colm swap → det changes sign.
- $k\begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} ka & kb \\ c & d \end{vmatrix}$ Scalars get multiplied with single row/single colm.
- can be broken row-wise $\begin{vmatrix} a+b & c+d \\ e & f \end{vmatrix} = \begin{vmatrix} a & c \\ e & f \end{vmatrix} + \begin{vmatrix} b & d \\ e & f \end{vmatrix}$
- For Dar or diagonal matrix → det = product of diagonal values.
- $R_i \leftarrow R_i \pm kR_j$ } Det does NOT change.
- Dependent rows ≡ dep. colms } det = 0
- Det(AB) = Det(A) · Det(B)
- Det(kA) = k^n Det(A)
- Det $A^{-1} = \frac{1}{\det A}$
- Det(A) = Det(A^T)